

AMAN KUMAR JHA

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Second-year B.Tech CSE (AI/ML) student with hands-on experience building full-stack web applications and production-grade AI systems. Developed a deployed movie recommendation engine processing 44,000+ records and an RAG-based PDF AI assistant with FastAPI and LangChain. Passionate about solving real-world problems at the intersection of machine learning and modern web development.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java

Frontend: React, Next.js, HTML5, CSS3, Responsive Web Design

Backend: Node.js, Express.js, FastAPI, REST APIs

AI / ML: Scikit-learn, TF-IDF, Cosine Similarity, LangChain, RAG, Vector Embeddings, Pandas, NumPy

Databases & Tools: FAISS, ChromaDB, Git, GitHub

Core CS: Data Structures & Algorithms, OOP, DBMS, Computer Networks, Operating Systems

PROJECTS

CineSync – Full Stack Movie Recommendation System | [React](#) · [Next.js](#) · [Python](#) · [Scikit-learn](#) · [TF-IDF](#) [\[GitHub ↗\]](#)

- ▶ Engineered a content-based recommendation engine on 44,000+ movie records using TF-IDF vectorization and Cosine Similarity, delivering sub-second personalized results.
- ▶ Built a full-stack application with a React/Next.js frontend and Python ML backend, integrated via RESTful APIs for seamless client-server data exchange.
- ▶ Designed a responsive, mobile-first UI with dynamic filtering and real-time recommendations, deployed to production for real-world usage.

Chat with PDF – RAG-Based AI Document Assistant | [Python](#) · [FastAPI](#) · [LangChain](#) · [FAISS/ChromaDB](#) · [Gemini API](#) [\[GitHub ↗\]](#)

- ▶ Developed an AI-powered document assistant enabling natural language querying of PDFs using Retrieval-Augmented Generation (RAG), eliminating manual document search.
- ▶ Engineered an end-to-end NLP pipeline — document ingestion, semantic chunking, embedding generation, and FAISS/ChromaDB vector indexing — for fast similarity retrieval.
- ▶ Built a high-performance FastAPI backend serving LLM-powered (Gemini/OpenAI) context-aware Q&A, supporting multi-turn conversational querying of uploaded documents.
- ▶ Designed an intuitive upload-and-chat interface enabling interactive, source-grounded responses across multi-page PDFs.

CERTIFICATIONS

Frontend Developer (React) — HackerRank | Verified | [View Certificate ↗](#)

Covers React component architecture, hooks, state management, CSS, and JavaScript fundamentals.

EDUCATION

B.Tech (Hons) – Computer Science Engineering (AI/ML)

2024 – 2028

SDGI Global University, Dasna, Ghaziabad, Uttar Pradesh

RELEVANT COURSEWORK

Data Structures & Algorithms • Object-Oriented Programming • Database Management Systems • Machine Learning • Computer Networks • Operating Systems